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Unit 6 - Expanding Knowledge on the Topic

PDF Documents

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UNIT 6 EXPANDING KNOWLEDGE ON THE TOPIC

The Story of Continental Drift and Plate Tectonics

EARLY THEORIES OF CONTINENTAL MOVEMENT

The story begins with the theory of continental drift, first proposed by meteorologist Alfred Wegener in 1912. Wegener noticed that the coastlines of Africa and South America seemed to fit together like pieces of a jigsaw puzzle. He suggested that the continents were once part of a massive supercontinent called Pangaea, which began to break apart during the Mesozoic era. His idea, however, was initially met with skepticism because he couldn't explain the mechanism that drove the continents apart.



DISCOVERY OF THE SEAFLOOR SPREADING

In the 1960s, the discovery of seafloor spreading at mid-ocean ridges provided the necessary mechanism for Wegener's theory. Scientists found that new ocean floor was being created at these ridges, pushing the seafloor and the continents apart.

UNDERSTANDING PLATE TECTONICS

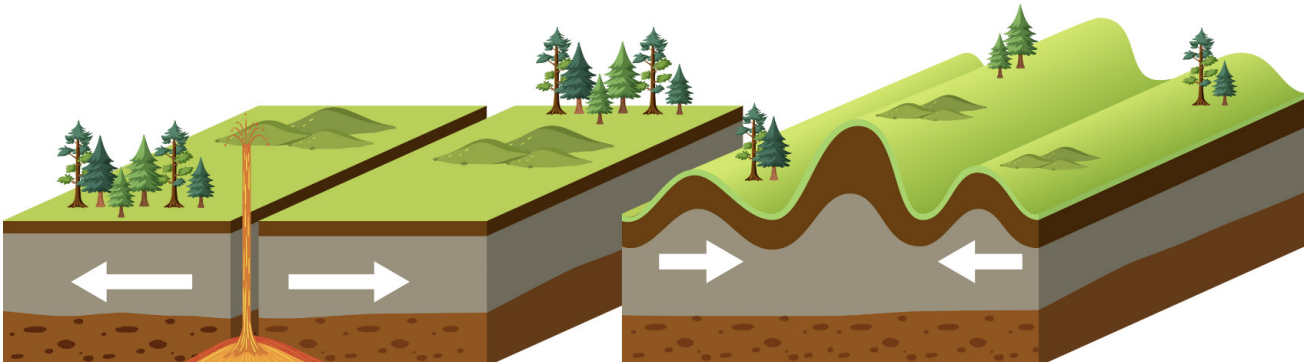


Plate tectonics is now understood to involve the Earth's lithosphere (the crust and the upper part of the mantle) broken into several large plates that float on the semi-fluid asthenosphere beneath them. These tectonic plates move very slowly, driven by the heat from the Earth's interior. Their interactions are responsible for not only the movement of continents but also most of the Earth's seismic activity, including earthquakes, volcanic eruptions, and the formation of mountain ranges.

MODERN IMPLICATIONS

Today, technologies like GPS allow scientists to measure the movement of tectonic plates in real-time. Understanding plate tectonics is crucial for assessing geological hazards, exploring resources, and studying environmental changes influenced by the configuration of continents and oceans.